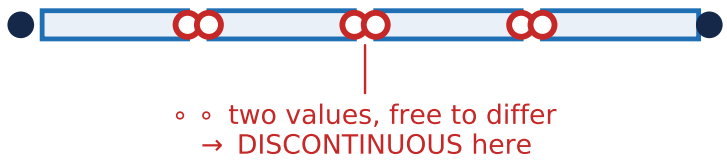
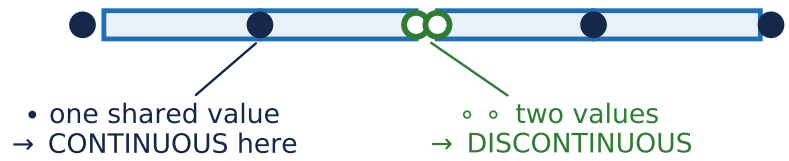
Case 9 --- DG for $-u'' = f$ in a bar
the exact answer is *smooth*

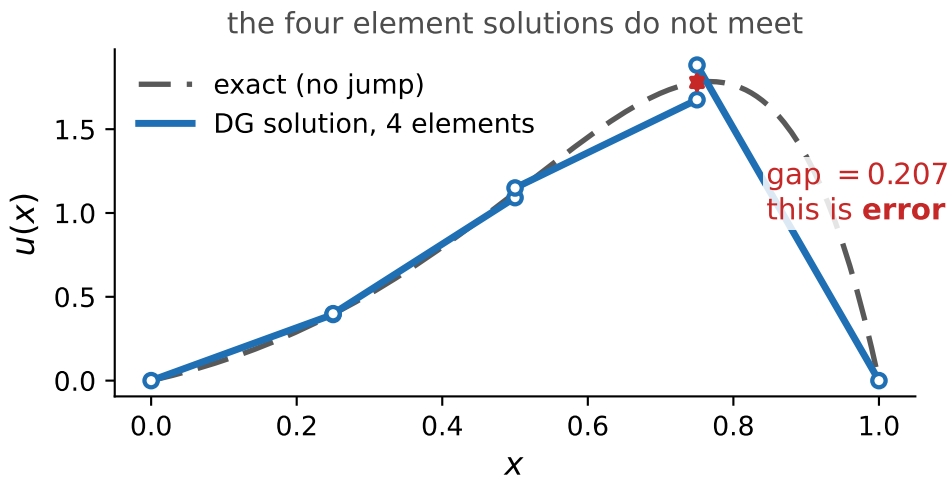


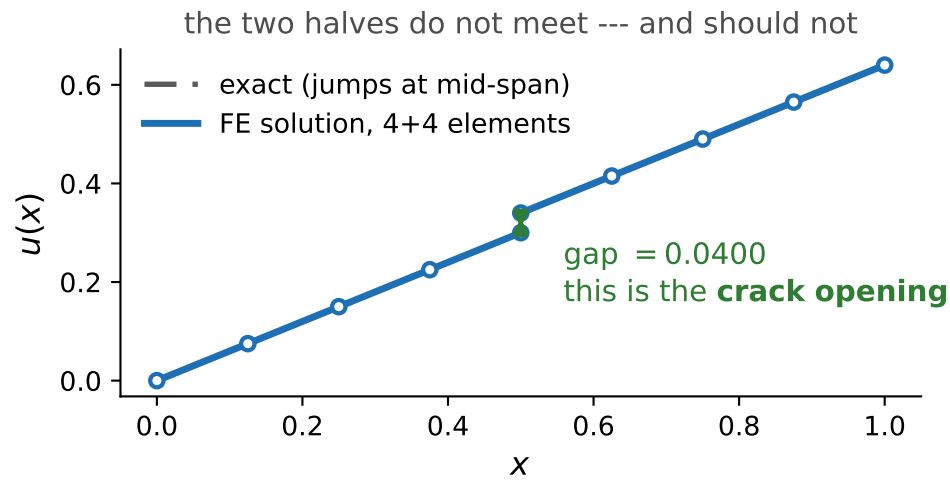
every face broken: each element owns its own nodes

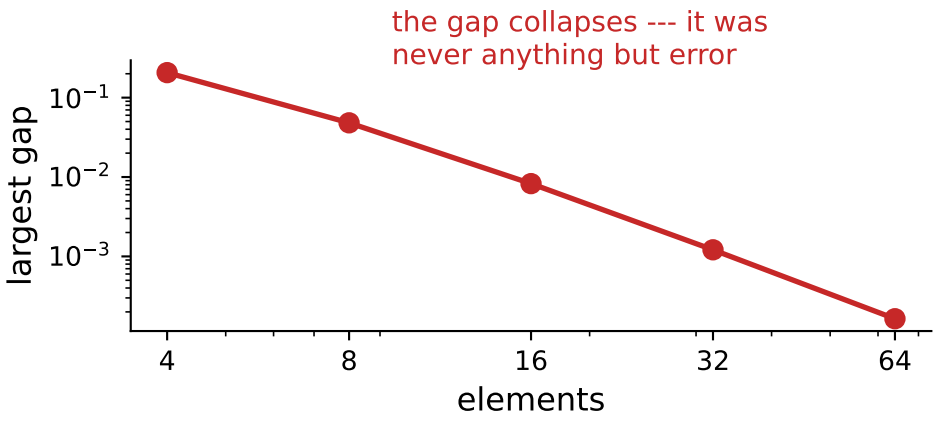
Case 10 --- cohesive zone in the same bar
the exact answer *really jumps*

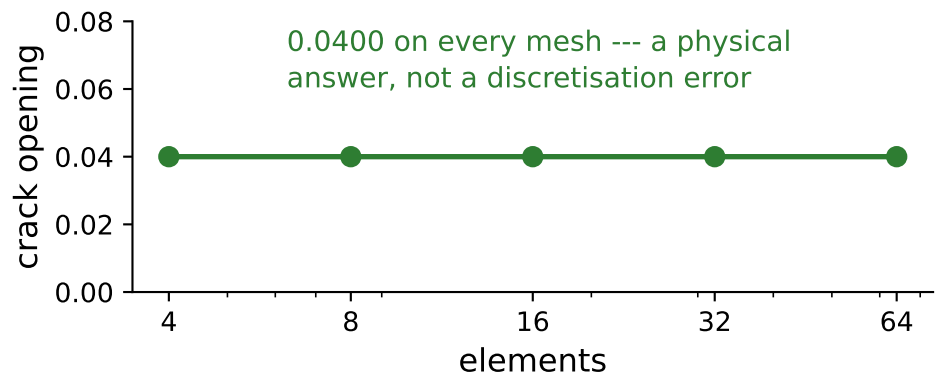


only the crack path broken; elsewhere the node is shared









DG ⇒ NON-CONFORMING
the face term is a penalty you tune; it exists only to suppress a jump the true answer never had

DG machinery ⇒ CONFORMING
the face term is a traction--separation law with a real G_c; the jump it allows is the one the true answer has